

Temporal Factorization of Factual Recall in Language Models

Overview

Factual recall in decoder-only language models involves at least two pieces of information: the **relation** being queried (e.g. *capital-of*) and the **entity** the relation applies to (e.g. *France*). This project asks a temporal question: do these two pieces of information become *generation-controlling at the final token* at the same depth, or at different depths?

The central finding is that recall is temporally factorised. Relation information becomes causally active at the final token position substantially earlier than entity information. Entity information is not absent early rather it is already strongly available at the entity token position from the first layers. However, its commitment to generation at the final token is deferred until late layers, after it is routed there. This pattern is referred to as **deferred entity commitment**.

Research question

When a language model recalls a fact, do relation and entity signals commit to the generation position at the same stage of computation? If they do not, factual recall should be understood as a multi-stage routing problem, not a single retrieval step.

Method

The primary technique is final token activation patching: at a chosen layer ℓ , the donor prompt’s hidden state at the final token position is inserted into the recipient’s final-token position, and the model’s first generated answer is checked to see whether it switches toward the donor-compatible answer. Sweeping ℓ across the network yields a transfer curve. Onset is defined as the first tested layer at which transfer exceeds 0.4 for two consecutive tested layers; threshold sensitivity (0.2–0.5) is checked in the appendix.

Two main conditions are compared:

- **Relation transfer:** donor and recipient share the entity but differ in relation. A successful patch means the final token state is being controlled by relation-type information.
- **Entity transfer:** same relation, different entity. A successful patch means the final token state is being controlled by entity specific answer information.

Experiments use four decoder-only models (Llama-3.2-3B, Llama-3-8B, Qwen2.5-3B, Phi-2) and eight prompt families spanning factual, morphological, lexical, and symbolic transformations (capital, language, past tense, present participle, plural, opposite, comparative, chemical symbol). Results are reported as pair-balanced relation curves and family-balanced entity curves to avoid record-count weighting artifacts.

Experiment 1: Transfer curves

In every model, relation transfer rises while entity transfer is still near zero, and relation onset precedes entity onset by 10–16 tested layers (31–44% of network depth):

Model	Rel. onset	Ent. onset	Gap	Depth
Llama-3.2-3B	L6	L18	12	43%
Llama-3-8B	L10	L20	10	31%
Qwen2.5-3B	L16	L32	16	44%
Phi-2	L14	L24	10	31%

The ordering holds in **all 16 model-threshold combinations** (thresholds 0.2–0.5). A wrong-entity control (patching a different entity of the same donor family) rules out donor-answer copying: at peak relation-transfer layers, relation-only transfer reaches 0.79–0.86 while donor-answer copying remains ≤ 0.11 .

Experiment 2: Both-change competition

When the donor differs in *both* relation and entity, outputs can be classified as relation-wins, entity-wins, original-retained, or mixed. Relation-wins dominate middle layers and entity-wins dominate late layers in every model, with the crossover layer tracking the entity onset from Experiment 1 to within ≤ 2 tested layers. Noise-patch controls produce ≤ 0.008 structured wins; self-patch preserves the original output at $\geq 99.2\%$; unrelated-donor patches yield high *late* entity-like overwrite but near-zero relation-wins (≤ 0.030), ruling out generic-patching artifacts as an explanation for mid-layer relation dominance.

Experiment 3: Entity token vs final token patching

The most direct test of the alternative explanation is “entity information is simply absent until late layers” and to patch at the *entity-token* position instead of the final-token position. The result is a clean double dissociation: entity-token patching succeeds at **90–100%** in early and middle layers, while final-token patching is at $\approx 2.4\%$ there. Late, the pattern reverses (final-token rises to 88–97%, entity-token collapses to 2–4%). The switch layer matches the entity onset from Experiments 1–2 in every model. *This supports the interpretation that deferred entity commitment reflects delayed routing rather than delayed knowledge.*

Experiment 4: Steering

As an independent check, relation steering directions are constructed as mean-difference vectors across same-entity/different-relation prompts, and entity steering directions as mean-difference vectors across same-relation/different-entity pairs. These are applied at mid- and late-layer zones. Relation directions steer effectively in middle layers; entity directions are substantially weaker mid-layer and strongest late. Random-direction baselines remain ≤ 0.015 , supporting direction-specificity.

Interpretation

Four complementary causal diagnostics which are transfer curves, both-change competition, entity-token vs final-token patching, and steering, all converge on the same transition layers in every model. Together they support a four-stage picture: (1) entity-token availability, (2) relation becomes final-token active, (3) entity becomes final-token committed, (4) late overwrite sensitivity. This separates two ideas that are routinely conflated in interpretability: *representing* information and *using* information to control generation.

Why this matters

If relation and entity signals become generation-controlling at different depths, then (i) layer-targeted interventions such as model editing or activation steering may affect relation-type vs answer-identity computation differently; (ii) decodability-based monitoring methods need to distinguish information *availability* from causal *commitment*; and (iii) similar staging may matter for chain-of-thought and multi-hop settings.

Limitations and next steps

The experiments use controlled fill-in-the-blank families and greedy first-answer generation; naturalistic QA, longer contexts, and free-form reasoning remain untested, and absolute layer numbers should not be interpreted as universal across the 3B–8B range. The experiments identify a robust causal timing asymmetry, not a complete circuit: the mechanism that routes entity information from the entity-token position to the final-token position is not yet localised. Natural follow-ups are: (a) localising the routing mechanism via attention-pattern analysis and head-level patching; (b) scaling tests as to see if the gap shrink, grow, or stay invariant with model size and instruction-tuning?; (c) extending the framework to multi-hop and chain-of-thought prompts, where multiple relation-entity bindings would have to be staged.

Artifact contents

This repository contains the project summary, main figures, processed result tables, and lightweight plotting code. It is a compact research artifact rather than the full experimental pipeline.